C5.1 Mapping coherency operations to snoop operations

This section describes the snoop transactions seen on the snoop address channel by a cached master that is being snooped by an initiating master.

When an initiating master issues a transaction, the interconnect is responsible for carrying out any snoop transactions required to complete the original transaction.

Not all transactions issued by an initiating master are permitted on the snoop address channel. Table C5-1 shows the protocol-recommended mappings between transactions issued by the initiating master and the snoop transactions seen on the snoop address channel by a cached master.

Table C5-1 Recommended transaction mappings

Transaction from initiating master	Transaction to snooped master
ReadNoSnoop	Not snooped
ReadOnce	ReadOnce
ReadClean	ReadClean
ReadNotSharedDirty	ReadNotSharedDirty
ReadShared	ReadShared
ReadUnique	ReadUnique
CleanUnique	CleanInvalid
MakeUnique	MakeInvalid
CleanShared	CleanShared
CleanInvalid	CleanInvalid
MakeInvalid	MakeInvalid
WriteNoSnoop	Not snooped
WriteUnique	CleanInvalid
WriteLineUnique	MakeInvalid
WriteBack	Not snooped
WriteClean	Not snooped
Evict	Not snooped

Note

The interconnect can use other mappings that force the same cache line state changes in a snooped master. See *Alternative snoop transactions* on page C5-211.

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C5.1.1 Permitted snoop transactions

Although the protocol does not require a fixed set of transaction mappings, the protocol does require that only the following defined subset of transactions are seen on the snoop address channel of a cached master:

- ReadOnce
- ReadClean
- ReadNotSharedDirty
- ReadShared
- ReadUnique
- CleanInvalid
- MakeInvalid
- CleanShared.

C5.1.2 Transactions not permitted as snoop transactions

The following transactions must not be seen on the snoop address channel of a cached master:

- ReadNoSnoop
- CleanUnique
- MakeUnique
- WriteNoSnoop
- WriteUnique
- WriteLineUnique
- WriteBack
- WriteClean
- Evict.

C5.1.3 Alternative snoop transactions

Table C5-2 shows each permitted snoop transaction on the snoop address channel, the required cache line state change for the transaction, and the alternative snoop transaction that can be used. For completeness, the snoop transaction option column includes the original snoop transaction.

Table C5-2 Snoop transaction options on the address snoop channel

Snoop transaction	Required cache line state change	Snoop transaction option
ReadOnce	None	ReadOnce ReadClean, ReadNotSharedDirty, ReadShared ReadUnique, CleanInvalid CleanShared
ReadClean	Shared or Invalid	ReadClean, ReadNotSharedDirty, ReadShared ReadUnique, CleanInvalid
ReadNotSharedDirty	Shared or Invalid	ReadClean, ReadNotSharedDirty, ReadShared ReadUnique, CleanInvalid
ReadShared	Shared or Invalid	ReadClean, ReadNotSharedDirty, ReadShared ReadUnique, CleanInvalid
ReadUnique	Invalid	ReadUnique, CleanInvalid
CleanInvalid	Invalid	ReadUnique, CleanInvalid
CleanShared	Clean or Invalid	ReadUnique, CleanInvalid CleanShared

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Mapping to different snoop transactions can simplify the design of a snooped master. For example, a snooped master can handle all snoop transactions in the same way as a ReadUnique transaction. This is permitted because a ReadUnique transaction, as Table C5-2 on page C5-211 shows, is an alternative snoop transaction for all other snoop transactions.

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